NAME:
SECTION:

DIRECTIONS: Solve for the following given problems.

1. **Estimate the gravitational force** between two sumo wrestlers, with masses 220 kg and 240 kg, when they are embraced and their centers are 1.2 m apart.
2. Given the gravitational force of $7.42 \times 10^{-8}\, N$ between a 5-kg spherical steel ball and another spherical steel ball separated by a center-to-center distance of 15 cm, **find the mass of the second spherical steel ball**.
3. The International Space Station (ISS) has an approximate mass of 370000 kg. During an activity, a suited astronaut with a total mass of 150 kg experiences a subtle gravitational pull of exactly $9.26 \times 10^{-6}\, N$ directed toward the station's core. Find how far the astronaut is from the space station?
4. A rock is dropped from a garage roof from rest. The roof is 6.0 m from the ground.
   a. Determine how long it takes the rock to hit the ground.
   b. Determine the velocity of the rock as it hits the ground.
   c. A second rock is projected straight upward from ground level at the moment the first rock was released. This second rock had an initial upward velocity of +6.0 m/s. How long will this second rock take to reach maximum height?
5. A stone is thrown straight down from a bridge. It hits the water 50 meters below with a final velocity of 35 m/s. Find the initial velocity.
6. Find the starting velocity of a package that is dropped/thrown downward from a helicopter. It travels a distance of 120 meters in 4 seconds before hitting the ground.
7. A laboratory technician, with a force of 45 N uses a specialized pneumatic launcher to shoot a tiny rubber stopper weighing 450 g during an experiment. Find the acceleration in m/s^2
8. A net force of 50 N acts on an object, causing its velocity to change from 2 m/s to 8 m/s over a time interval of 3 s. Find the mass of the object.
9. A crate is being pushed across a frictionless surface with a horizontal force of 50 N. If the create accelerates at $2.5\, m/s^2$, what is the weight of the crate?
10. A box is being pushed with a force of 100 N across the floor. If the box has a weight of 490 N, what is the acceleration of the box?
11. A 1,200 kg car accelerates from rest to a velocity of 25 m/s over a time period of 5 s. What net force is required to cause the acceleration?
12. Given an object with a weight of 490 N, an acceleration of $4.5\, m/s^2$, find the net force of the object?
13. A bike is moving at the rate of 30 m/s carrying a momentum of 5000 kg m/s. Determine its mass.
14. A 150000-g body (A) is travelling at a velocity of 10 m/s, while a 200-kg body (B) is moving beside it at twice the velocity of A.
    a. What is the momentum of the A?
    b. What is the momentum of the B?
    c. Which of the two has a higher momentum?
15. If a 2 kg object has a momentum of 10 kg m/s and moves at that constant speed for 4 seconds, how far does it travel?
16. A 0.5 kg football is kicked straight toward a player. Just before hitting the player's chest, the ball has an initial momentum of 6 kg m/s to the east. After bouncing straight back off the chest, its final momentum is 4 kg m/s to the west. What is the impulse delivered to the ball by the player's chest?
17. A tennis ball is hit with an initial momentum of 15 kg m/s to the left. It is struck by a racket and returns with a final momentum of 20 kg m/s in the opposite direction. What is the impulse?

18. You are a playing soccer. A soccer ball comes toward you with an initial momentum of 12 kg m/s. You kick it in the opposite direction giving it a final momentum of -20 kg m/s. Your foot was in contact with the ball for 0.04 seconds. Find the average force your foot exerted on the ball.
19. A 5 kg motorized skateboard is cruising down a smooth path. The rider hits the regenerative brakes for 4 seconds, causing the skateboard to slow down to a final velocity of 2 m/s. During these 4 seconds of braking, the skateboard experiences a loss in momentum of -40 kg m/s.

ANSWER KEY

1. $F = 2.45 \times 10^{-6}\ N$
2. $M_2 \approx 5\ kg$
3. $d = 19.99\ m$
4. a. $1.11\ s = t$

   b. $10.88\ m/s$

   c. $0.61\ s = t$
5. $15.65 \frac{m}{s} = v_i$
6. $10.4 \frac{m}{s} = v_i$
7. $a = 100\ m/s^2$
8. $25\ kg = m$
9. $196\ N = W$
10. $2 \frac{m}{s^2} = a$
11. $F = 6000\ N$
12. $F = 225\ N$
13. $166.67\ kg = m$
14. a. $p = 1500\ kg\ m/s$

    b. $p = 4000\ kg\ m/s$

    c. B has a higher momentum than A.
15. $20\ m = d$
16. $\vec{J} = -10\ kg \frac{m}{s}$
17. $\vec{J} = 35\ kg \frac{m}{s}$
18. $-800\ N = F$
19. $10 \frac{m}{s} = v_i$

"For the LORD gives wisdom, from his mouth come knowledge and understanding;"

Proverbs 2:6